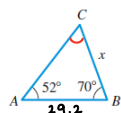


- 5.5
1) Use law of sines to find indicated side x



$$\angle C = 180^\circ - 52^\circ - 70^\circ = 58^\circ$$

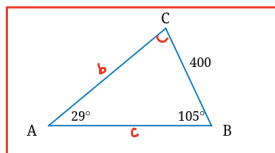
$$\frac{x}{\sin(52^\circ)} = \frac{29.2}{\sin(58^\circ)}$$

$$x = \frac{(29.2 \sin(52^\circ))}{\sin(58^\circ)}$$

$$x \approx 27.1 \text{ un}$$

- 2) sketch the triangle

$$\angle A = 29^\circ, \angle B = 105^\circ, a = 400$$



Solve it using law of sines

$$\angle C = 180^\circ - 105^\circ - 29^\circ = 46^\circ$$

$$\frac{b}{\sin(105^\circ)} = \frac{400}{\sin(29^\circ)}$$

$$b = \frac{(400 \sin(105^\circ))}{\sin(29^\circ)}$$

$$b \approx 797.0 \text{ un}$$

$$\frac{c}{\sin(46^\circ)} = \frac{400}{\sin(29^\circ)}$$

$$c = \frac{(400 \sin(46^\circ))}{\sin(29^\circ)}$$

$$c \approx 593.5 \text{ un}$$

- 3) Use Law of Sines to solve for all possible Δ s that satisfy given conditions.

$$a = 107, b = 83, \angle A = 138^\circ$$

$$\text{SSA}, A > 90^\circ$$

$$a > b \rightarrow 1 \text{ solution}$$

$$\frac{\sin(138^\circ)}{107} = \frac{\sin B}{83}$$

$$\angle B = \sin^{-1} \left[\frac{83 \sin(138^\circ)}{107} \right]$$

$$\angle B \approx 31.3^\circ$$

$$\angle C = 180^\circ - 138^\circ - \angle B$$

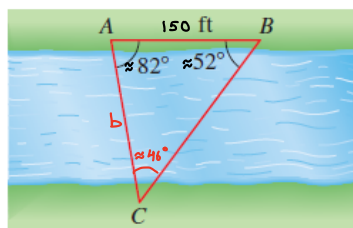
$$\angle C \approx 10.7^\circ$$

$$\frac{107}{\sin(138^\circ)} = \frac{c}{\sin C}$$

$$c = \left[\frac{107 \sin C}{\sin(138^\circ)} \right]$$

$$c \approx 29.8 \text{ un}$$

- 4) approximate distance from A to C

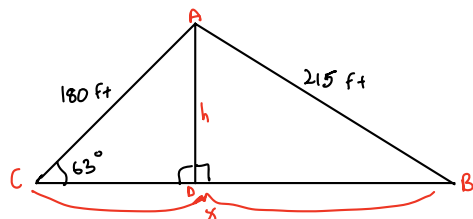


$$\frac{b}{\sin(52^\circ)} = \frac{150}{\sin(46^\circ)}$$

$$b = \frac{150 \sin(52^\circ)}{\sin(46^\circ)}$$

$$b \approx 164 \text{ feet}$$

- 5) Short-wave radio antenna is supported by two guy wires, 180 ft and 215 ft long each wire is attached to top of antenna and anchored to ground at two anchor points on opposite sides of antenna. Shorter wire makes angle of 63° with ground. How far apart are the anchor points?



$$\sin 63^\circ = \frac{h}{180}$$

$$h = 180 \sin 63^\circ$$

$$BD^2 + h^2 = 215^2$$

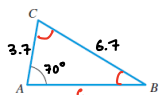
$$BD = \sqrt{215^2 - h^2}$$

$$CD^2 + h^2 = 180^2$$

$$CD = \sqrt{180^2 - h^2}$$

$$\begin{aligned} x &= CD + BD \\ x &\approx 225 \text{ ft} \end{aligned}$$

- 6) Solve the triangle using Law of Sines



$$\frac{\sin(70^\circ)}{6.7} = \frac{\sin B}{3.7}$$

$$B = \sin^{-1} \left[\frac{3.7 \sin(70^\circ)}{6.7} \right]$$

$$\angle B \approx 31.3^\circ$$

$$\angle C = 180^\circ - 70^\circ - \angle B$$

$$\angle C \approx 78.4^\circ$$

$$\frac{6.7}{\sin(70^\circ)} = \frac{c}{\sin C}$$

$$c = \frac{6.7 \sin C}{\sin(70^\circ)}$$

$$c \approx 7.0 \text{ in}$$

- 7) Use Law of Sines to solve for all possible $\triangle$ s that satisfy given conditions.

$$a = 73, b = 105, \angle A = 23^\circ$$

$$\text{SSA}, A < 90^\circ$$

$$b \sin A \approx 41^\circ$$

$$b > a > b \sin A \rightarrow 2 \text{ solutions}$$

$$\frac{73}{\sin(23^\circ)} = \frac{105}{\sin(B_1)}$$

$$\theta_1 = \sin^{-1} \left[\frac{105 \sin(23^\circ)}{73} \right]$$

$$\angle B_1 \approx 34.2^\circ$$

$$C_1 = 180^\circ - 23^\circ - B_1$$

$$\angle C_1 \approx 122.8^\circ$$

$$\frac{73}{\sin(23^\circ)} = \frac{c_1}{\sin(C_1)}$$

$$c_1 = \frac{73 \sin(C_1)}{\sin(23^\circ)}$$

$$c_1 \approx 157.0 \text{ in}$$

$$\theta_2 = 180^\circ - B_1$$

$$\angle B_2 \approx 145.8^\circ$$

$$C_2 = 180^\circ - 23^\circ - B_2$$

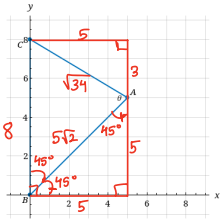
$$\angle C_2 \approx 11.2^\circ$$

$$\frac{73}{\sin(23^\circ)} = \frac{c_2}{\sin C_2}$$

$$c_2 = \frac{73 \sin C_2}{\sin(23^\circ)}$$

$$c_2 \approx 36.3^\circ$$

8) Use law of Sines to find θ

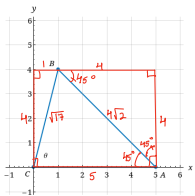


$$\frac{\sin 45^\circ}{\sqrt{34}} = \frac{\sin \theta}{8}$$

$$\theta = \sin^{-1} \left[\frac{8 \sin(45^\circ)}{\sqrt{34}} \right]$$

$$\theta \approx 76.0^\circ$$

9) Use law of Sines to find θ

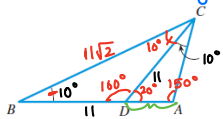


$$\frac{\sin(45^\circ)}{\sqrt{17}} = \frac{\sin \theta}{4\sqrt{2}}$$

$$\theta = \sin^{-1} \left[\frac{4\sqrt{2} \sin(45^\circ)}{\sqrt{17}} \right]$$

$$\theta \approx 76^\circ$$

10) Complete triangle



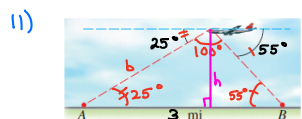
$$\angle DCB = 10^\circ$$

$$\angle A = 150^\circ$$

$$\frac{AD}{\sin(10^\circ)} = \frac{11}{\sin(150^\circ)}$$

$$AD = \frac{11 \sin(10^\circ)}{\sin(150^\circ)}$$

$$AD \approx 3.82 \text{ un}$$



(a) Find distance of plane from Point A

$$\frac{b}{\sin(55^\circ)} = \frac{3}{\sin(100^\circ)}$$

$$b = \frac{3 \sin(55^\circ)}{\sin(100^\circ)}$$

$$b \approx 2.50 \text{ mi}$$

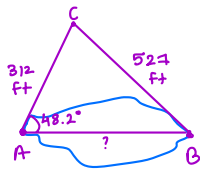
(b) Find elevation of plane

$$\sin(25^\circ) = \frac{h}{b}$$

$$h = b \sin(25^\circ)$$

$$h \approx 1.05 \text{ mi}$$

- 12) Points A and B are separated by a lake. To find distance between them, surveyor locates point C on land such that $\angle CAB = 48.2^\circ$ and then measures CA as 312 ft and CB as 527 ft. Find distance between A and B.



$$\frac{\sin(48.2)}{527} = \frac{\sin B}{312}$$

$$\angle B = \sin^{-1} \left[\frac{312 \sin(48.2^\circ)}{527} \right]$$

$$\angle C = 180^\circ - 48.2^\circ - \angle B$$

$$\frac{527}{\sin(48.2^\circ)} = \frac{AB}{\sin(C)}$$

$$AB = \frac{527 \sin(C)}{\sin(48.2^\circ)}$$

$$AB \approx 680.86 \text{ ft}$$